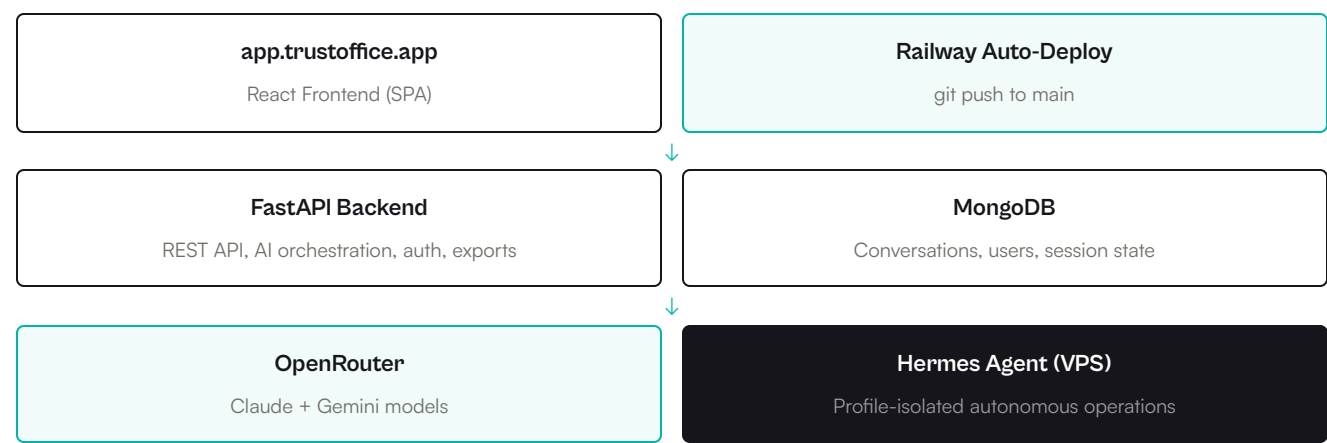


TrustOffice Platform Briefing

Prepared for meeting with Jake • July 15, 2026 • Jeff Kohler

01 Infrastructure

Fully owned stack. No shared SaaS backends. Every component is ours, deployed from our own repositories.



Component	Technology	Role
Hosting	VPS (Linux)	Hermes Agent, monitoring, scheduled tasks
Deployment	Railway	Auto-deploys on git push. No manual steps.
Backend	FastAPI (Python)	REST API, AI orchestration, business logic
Frontend	React (SPA)	Web chat portal at app.trustoffice.app
Database	MongoDB	Conversations, user accounts, session state
AI Layer	OpenRouter	Routes to Claude (complex) and Gemini (fast)
Agent	Hermes (isolated profile)	Monitoring, maintenance, content, cron jobs

02 Security

Layer	Control	Status
Transport	TLS/HTTPS on all endpoints. No plaintext.	Live
CORS	Locked to trustoffice.app + app.trustoffice.app only.	Live
User Auth	JWT, signed and time-limited, validated per request.	Live
Admin Access	Separate API key. User JWTs cannot escalate.	Live
Input Validation	Pydantic models on all endpoints. No raw input.	Live
Profile Isolation	Hermes profiles are sandboxed. No cross-profile data access.	Live
Secrets	Local env vars on VPS/Railway. Never in code or git.	Live
Data Scoping	User data scoped per account. No global read access.	Live

03 Performance Fixes (Deployed Tonight)

Both issues Jake reported are **fixed and live**.

Issue	Root Cause	Fix	Result
2-3 min response times	Sync AI calls blocked the event loop. Sequential round-trips before first token.	asyncio.to_thread, parallelized intent classification, reduced prompt from 15K to 8K tokens, timeout 90s to 30s.	15-20s
Mobile drops on backgrounding	No visibility handler. OS kills fetch connection when tab is backgrounded.	Wake Lock API, visibilitychange handler, auto-reconnection, polling fallback, partial response saving.	Fixed

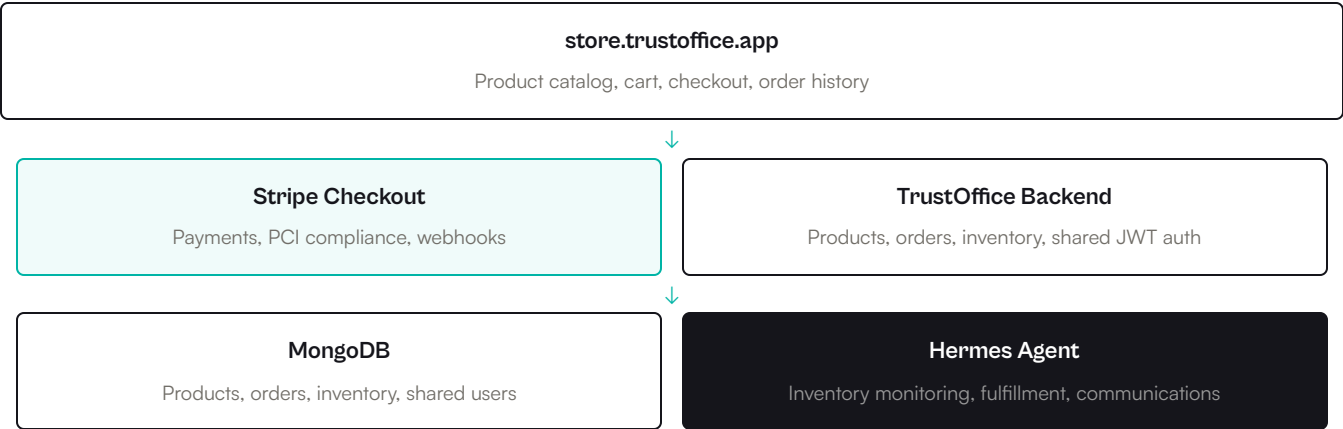
04 Feature Roadmap

Feature	Approach	Effort
Image upload in chat	Multipart upload endpoint, Cloudflare R2 storage, signed URLs, inline thumbnails in conversation.	1-2 days
CSV report downloads	Backend already has export patterns. Add /reports/export endpoint, query MongoDB, serve as file download.	1-2 days
PDF generation	WeasyPrint (HTML/CSS to PDF). Branded templates. /reports/pdf endpoint with same query params as CSV.	~1 week
Email notifications	Notification triggers in backend, email via Brevo/SendGrid, user preferences for opt-in/out.	~1 week
In-app real-time notifications	WebSocket per active session. Server pushes events (report ready, task done).	~2 weeks

Recommended priority: CSV export and file upload first (both live within a week). PDF generation and email notifications next. Real-time in-app notifications last.

05 E-Commerce Vision

Separate storefront at store.trustoffice.app. Stripe Checkout for payments (PCI-compliant, no card data on our servers). Hermes manages inventory, fulfillment, and customer communications autonomously.



Inherits all existing security: TLS, JWT, CORS, profile isolation. No card data stored. ~3-5 week build for MVP with automated operations.

06 Monitoring & Reliability

Three new systems went live tonight. We now catch issues before users do.

System	What It Does	Status
Synthetic Canary	Hits AI chat endpoint every 30 min. Measures time-to-first-token. Alerts if >5s (warning) or >10s (critical).	Live
Post-Deploy Smoke Test	9 endpoint checks after every deploy: health, auth, CORS, AI providers, bundle hash, polling endpoint.	Live
In-App Error Reporting	Frontend captures JS crashes and React render errors. Backend logs all unhandled exceptions. Queryable via admin API.	Live
Weekly Audit	5 new checks: async safety scan, AI response time, mobile resilience, timeout verification, prompt size.	Live

Summary

Topic	Key Takeaway
Infrastructure	Fully owned stack. Profile-isolated agent. No shared SaaS.
Security	TLS, CORS, JWT, admin API key. Profile isolation. Per-user data scoping.
Performance	Both issues fixed and live. Responses now 15-20s. Mobile survives backgrounding.
Features	CSV export and file upload: 1-2 days each. PDF and notifications: ~1 week each.
E-Commerce	Stripe Checkout + Hermes operations. ~3-5 week build.
Monitoring	Canary, smoke test, error reporting, weekly audit. All live.